

StockPilot SaaS - Pro Technical Documentation (v2)

1. Product Overview

StockPilot is a SaaS inventory & production management system designed for small-to-medium businesses. It tracks stock movements, production, dispatch, and provides real-time business insights.

2. System Architecture

Frontend: Next.js (App Router, Client Components)
Backend: Supabase (PostgreSQL + API)
Auth: Supabase Auth (future-ready)
State: React Hooks
Styling: Tailwind CSS

3. Data Flow

User Action → Next.js UI → Supabase Query → Database
← Processed Data ← Business Logic Layer

4. Project Structure

```
src/
├── app/
│   ├── components/
│   │   ├── Navbar.js
│   │   └── Logo.js
│   ├── dashboard/page.js
│   └── layout.js
├── lib/
│   ├── supabase.js
│   └── stock.js
├── config/
└── company.js
```

5. Database Schema

Products Table

```
products
id (PK)
name
minimum_stock
cost_price
selling_price
created_at
```

Production Table

```
production
id (PK)
product_id (FK)
quantity
created_at
```

Dispatch Table

```
dispatch
id (PK)
product_id (FK)
customer_id (FK)
quantity
created_at
```

6. Relationships

```
products 1 → many production
products 1 → many dispatch
customers 1 → many dispatch
```

7. Stock Engine

```
Current Stock =
Total Production - Total Dispatch

export async function getCurrentStock(product){
  const { data: production } = await supabase
    .from("production").select("quantity")
    .eq("product_id", product.id);

  const { data: dispatch } = await supabase
    .from("dispatch").select("quantity")
    .eq("product_id", product.id);

  return production.reduce((s,x)=>s+x.quantity,0)
    - dispatch.reduce((s,x)=>s+x.quantity,0);
}
```

8. Layout System

```
<body className="flex">
  <Navbar />
  <main className="ml-64 p-6 w-full">
    {children}
  </main>
</body>
```

9. Navbar System

Responsive sidebar with collapse + hover expand behavior.

- Expanded: shows logo + labels

- Collapsed: shows icons only
- Hover: temporary expand

10. Dashboard Module

Displays business KPIs and operational insights.

Metrics:

- Total Products
- Current Stock
- Today's Production
- Today's Dispatch

Additional Features:

- Low Stock Alerts
- Recent Activity Timeline
- Scalable for charts integration

11. Business Rules

- Stock must never go negative (future validation)
- Minimum stock triggers alerts
- Every movement must be logged

12. Scalability Plan

Phase 1: Basic CRUD + Dashboard
Phase 2: Analytics + Charts
Phase 3: Multi-user roles
Phase 4: Multi-tenant SaaS

13. Security Considerations

- Row Level Security (Supabase)
- Auth-based access control
- API validation

14. Future Features

- Role-based access
- Financial analytics
- Supplier module
- Purchase orders
- Mobile responsiveness improvements

15. Summary

StockPilot is structured to evolve from a simple stock tracker into a full ERP-grade SaaS platform with scalable architecture and clean separation of concerns.